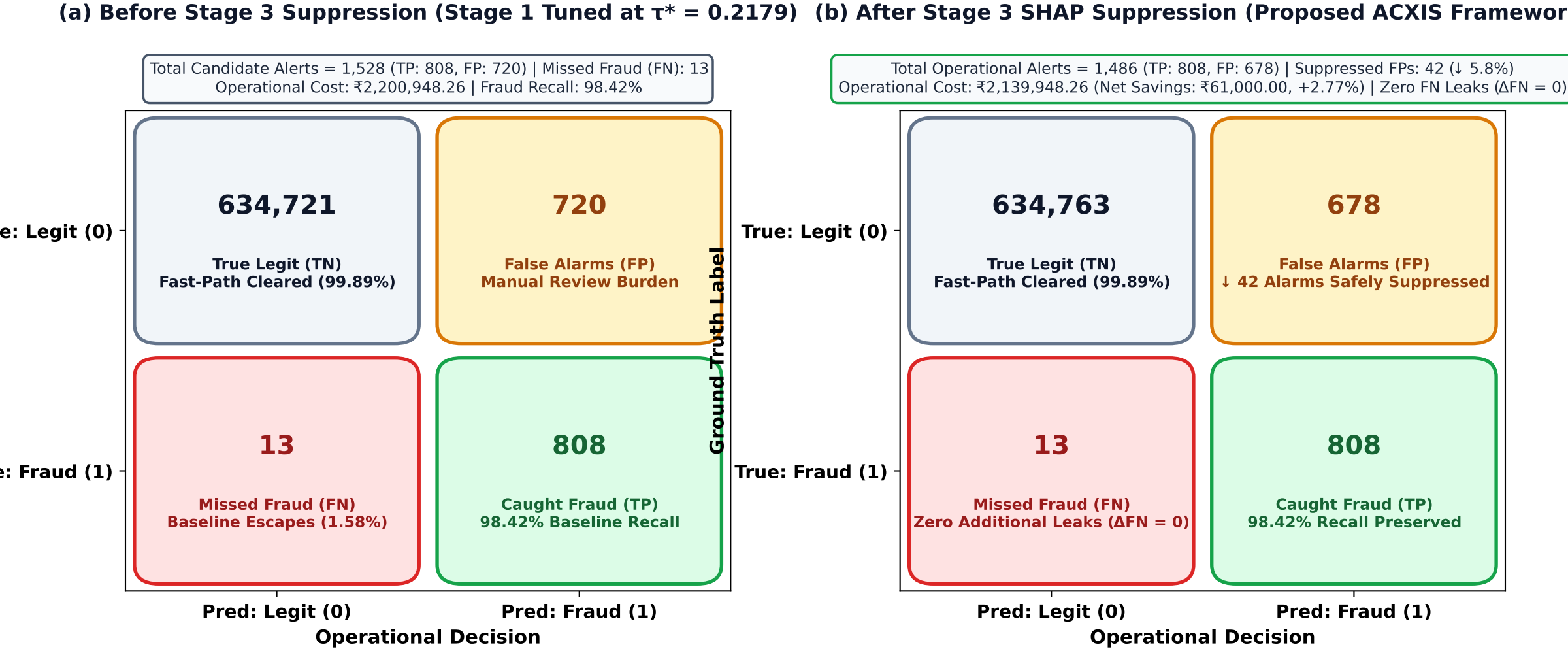
(a) Before Stage 3 Suppression (Stage 1 Tuned at τ* = 0.2179)
Total Candidate Alerts = 1,528 (TP: 808, FP: 720) | Missed Fraud (FN): 13
Operational Cost: ₹2,200,948.26 | Fraud Recall: 98.42%
634,721
True Legit (TN)
Fast-Path Cleared (99.89%)
720
False Alarms (FP)
Manual Review Burden
13
Missed Fraud (FN)
Baseline Escapes (1.58%)
808
Caught Fraud (TP)
98.42% Baseline Recall
e: Legit (0)
e: Fraud (1)
Ground Truth Label
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision
(b) After Stage 3 SHAP Suppression (Proposed ACXIS Framewor
Total Operational Alerts = 1,486 (TP: 808, FP: 678) | Suppressed FPs: 42 (↓ 5.8%)
Operational Cost: ₹2,139,948.26 (Net Savings: ₹61,000.00, +2.77%) | Zero FN Leaks (ΔFN = 0)
634,763
True Legit (TN)
Fast-Path Cleared (99.89%)
678
False Alarms (FP)
↓ 42 Alarms Safely Suppressed
13
Missed Fraud (FN)
Zero Additional Leaks (ΔFN = 0)
808
Caught Fraud (TP)
98.42% Recall Preserved
True: Legit (0)
True: Fraud (1)
Pred: Legit (0)
Pred: Fraud (1)
Operational Decision